

การตั้งค่าตัวตนและการเชื่อมต่อความปลอดภัย (Setup & Authentication)

- **ตั้งค่า Git Config:** เริ่มต้นด้วยการระบุตัวตนบนเครื่องคอมพิวเตอร์ โดยตั้งชื่อ Username เป็น tankeiei และอีเมลนักศึกษา tankun.p@kkumail.com เพื่อใช้เป็นข้อมูลในการบันทึกประวัติการ Commit
- **สร้างกุญแจความปลอดภัย (SSH Key):** ใช้คำสั่ง ssh-keygen -t ed25519 -C "tankun.p@kkumail.com" เพื่อสร้างคู่กุญแจดิจิทัล (Public/Private Key) สำหรับใช้เชื่อมต่อไปยัง GitHub อย่างปลอดภัยโดยไม่ต้องกรอกรหัสผ่านบ่อยๆ

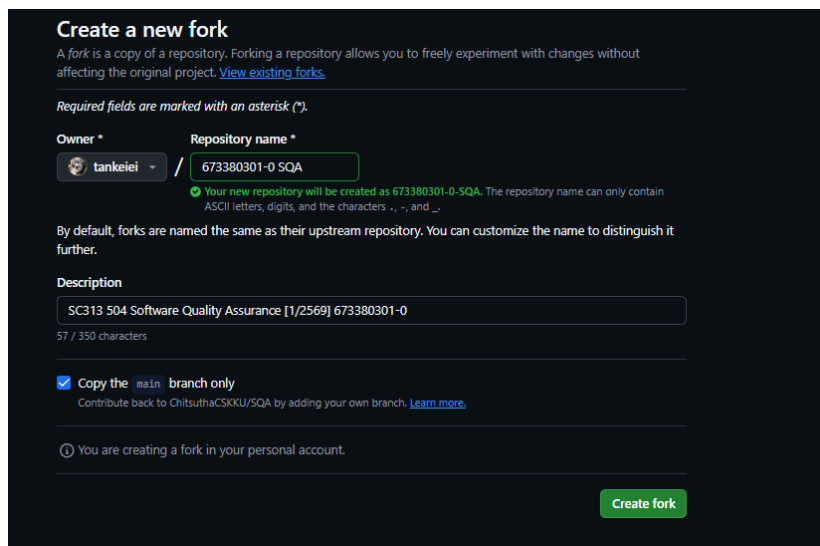
```
filter.its.required=true
user.name=tankeiei
user.email=tankun.p@kkumail.com
~
```

```
tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA
$ ssh-keygen -t ed25519 -C "tankun.p@kkumail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/tanku/.ssh/id_ed25519):
Created directory '/c/Users/tanku/.ssh'.
Enter passphrase for "/c/Users/tanku/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/tanku/.ssh/id_ed25519
Your public key has been saved in /c/Users/tanku/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:jg676KP/USdXaBAMZPqUlg2m3/GEXytEaTRWDanj14g tankun.p@kkumail.com
The key's randomart image is:
+--[ED25519 256]--+
|  .==O. .+OO+   |
|  +..= . O+.. .  |
|  . O+  O+..    |
|  O.   .O.= .   |
|  ..O.SO + o    |
|  ..*E * +      |
|  O . . O       |
|  .. =          |
|  .++O+..       |
+-----[SHA256]-----+

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA
$ |
```

การสร้างคลั่งและดึงโค้ดลงเครื่องคอมพิวเตอร์ (Fork & Clone)

- **ทำการ Fork โปรเจกต์:** กดทำสำเนาโปรเจกต์วิชา SQA จากคลั่งของอาจารย์ (ChitsuthaCSKKU/SQA) เข้ามาเป็นของตัวเองในชื่อบัญชี tankeiei/673380301-0-SQA
- **Clone โค้ดลงเครื่อง:** ใช้คำสั่ง `git clone [https://github.com/tankeiei/673380301-0-SQA](https://github.com/tankeiei/673380301-0-SQA)` ดึงไฟล์ทั้งหมดจากคลั่งส่วนตัวบนหน้าเว็บลงมาที่เครื่องคอมพิวเตอร์ และย้ายโฟลเดอร์ทำงานเข้าไปด้วยคำสั่ง `cd 673380301-0-SQA/`
- **สลับกิ่งทำงาน (Switch Branch):** ใช้คำสั่ง `git switch 2026` เพื่อเปลี่ยนจากกิ่งหลัก (main) ไปทำงานบนกิ่ง 2026 ตามโครงสร้างที่กำหนดไว้



Create a new fork

A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. [View existing forks.](#)

Required fields are marked with an asterisk (*).

Owner * tankeiei / Repository name * 673380301-0-SQA

✔ Your new repository will be created as 673380301-0-SQA. The repository name can only contain ASCII letters, digits, and the characters -, ., and _.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description SC313 504 Software Quality Assurance [1/2569] 673380301-0

57 / 350 characters

☒ Copy the main branch only

Contribute back to ChitsuthaCSKKU/SQA by adding your own branch. [Learn more.](#)

📘 You are creating a fork in your personal account.

[Create fork](#)

```
tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA
$ git clone ^[[200~
fatal: repository '^[200~' does not exist

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA
$ git clone https://github.com/tankeiei/673380301-0-SQA
Cloning into '673380301-0-SQA'...
remote: Enumerating objects: 1706, done.
remote: Counting objects: 100% (374/374), done.
remote: Compressing objects: 100% (176/176), done.
remote: Total 1706 (delta 226), reused 257 (delta 184), pack-reused 1332 (from 1)
Receiving objects: 100% (1706/1706), 18.07 MiB | 4.26 MiB/s, done.
Resolving deltas: 100% (550/550), done.

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA
$ cd 673380301-0-SQA/

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (main)
$ git switch
2022 2023 2024 2025 2026 HEAD main

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (main)
$ git switch 2026
branch '2026' set up to track 'origin/2026'.
Switched to a new branch '2026'

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ |
```

การบันทึกและอัปเดตงานครั้งแรก (First Commit & Push)

- **เตรียมและบันทึกไฟล์:** ใช้คำสั่ง `git add .` เพื่อติดตามและเตรียมไฟล์ทั้งหมดที่เปลี่ยนแปลงในโปรเจกต์ จากนั้นทำจุดบันทึกเวอร์ชันด้วยคำสั่ง `git commit -m "เพิ่ม pom และสร้าง src ใหม่ สำหรับ lab3 เพิ่ม test 1 ตัว"`
- **ส่งโค้ดขึ้นระบบ:** ใช้คำสั่ง `git push origin 2026` ดันประวัติและไฟล์เวอร์ชันแรกนี้ขึ้นไปเก็บไว้บนคลังออนไลน์ของตัวเองบน GitHub

```

C:\Program Files\Git\bin>git add .
warning: in the working copy of '.idea/workspace.xml', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.log', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.core.runtime/.apt_generated/org.eclipse.core.runtime.log', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.m2e.logback/logback-2.7.101.20251017-1242.xml', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserSupport/default.browser.preferences', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserSupport/throttle_store.dat', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/SessionStorageRules/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Extension Rules/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Extension State/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Extension Scripts/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Extension Scripts/Log', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Extension State/Log', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Local Storage/LevelDb/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Local Storage/LevelDb/LOG', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Local Storage/LevelDb/LOG.old', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Session Storage/Log', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Session Storage/LOG.old', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Site Characteristics Database/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Site Characteristics Database/LOG.old', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Sync Data/LevelDb/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/Sync Data/LevelDb/LOG', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/favorites_diagnostic.log', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/shared_proto_db/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/shared_proto_db/metadata/LOG', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/shared_proto_db/metadata/LOG.old', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/shared_proto_db/metadata/CURRENT', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/shared_proto_db/metadata/LOG', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.metadata/.plugins/org.eclipse.swt/EclipseWebBrowserDefault/shared_proto_db/metadata/LOG.old', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.labAssignment/Lab3_UnitTest/pom.xml', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.labAssignment/Lab3_UnitTest/src/main/java/com/lab3/Tab/ShiftCipher.java', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.labAssignment/Lab3_UnitTest/src/test/java/sqa/Tab/UnitCipher.java', LF will be replaced by CRLF the next time git touches it
warning: in the working copy of '.labAssignment/Lab3_UnitTest/target/maven-stamp/maven-complier-plugin/compile/default-compile/createFiles.lst', LF will be replaced by CRLF the next time git touches it

```

การบันทึกงานรอบสัปดาห์หลังแก้ไขเสร็จ (Final Commit & Push)

- **อัปเดตโค้ดเข้า Git:** หลังจากปรับปรุงชุดข้อมูลทดสอบจนเสร็จสมบูรณ์แล้ว ใช้คำสั่ง `git status` เพื่อรีเช็คไฟล์ และใช้ `git add .` ตรวจสอบไฟล์คลาส `UnitTestCipher.java` เข้าสู่สถานะเตรียมส่ง
- **คอมมิทรอบสุดท้าย:** สั่งบันทึกเวอร์ชันจบงานด้วยคำสั่ง `git commit -m "เพิ่ม test case ให้ครบ 5"`
- **ดันงานขึ้นหน้าเว็บ:** ปิดท้ายด้วยคำสั่ง `git push` เพื่ออัปเดตโค้ดทดสอบที่สมบูรณ์ทั้งหมดขึ้นไปเก็บบนกิ่ง 2026 บนหน้าโปรไฟล์ GitHub ของตนเอง เป็นอันเสร็จสิ้นกระบวนการทำงาน

```
tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git commit -m"เพิ่ม pom และสร้าง src ใหม่ สำหรับ lab3 เพิ่ม test 1 ตัว"
[2026 55f49cc] เพิ่ม pom และสร้าง src ใหม่ สำหรับ lab3 เพิ่ม test 1 ตัว
304 files changed, 17818 insertions(+), 17 deletions(-)
```

```

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git push origin 2026
Enumerating objects: 333, done.
Counting objects: 100% (333/333), done.
Delta compression using up to 16 threads
Compressing objects: 100% (233/233), done.
Writing objects: 100% (312/312), 8.94 MiB | 3.54 MiB/s, done.
Total 312 (delta 54), reused 25 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (54/54), completed with 3 local objects.
To https://github.com/tankeiei/673380301-0-SQA.git
   301331a..55f49cc 2026 -> 2026

```

```

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git status
On branch 2026
Your branch is up to date with 'origin/2026'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    LabAssignment/Lab3_UnitTest/ShiftCipher.java
        modified:   LabDemo/Lab3_Unit/junit6-labdemo/.project
        deleted:   LabDemo/Lab3_Unit/junit6-labdemo/target/classes/META-INF/MANIFEST.MF
        deleted:   LabDemo/Lab3_Unit/junit6-labdemo/target/classes/META-INF/maven/com.example/junit6-labdemo/pom.properties
        deleted:   LabDemo/Lab3_Unit/junit6-labdemo/target/classes/META-INF/maven/com.example/junit6-labdemo/pom.xml
        modified:  LabDemo/Lab3_Unit/junit6-labdemo/target/classes/com/example/Calculator.class
        modified:  LabDemo/Lab3_Unit/junit6-labdemo/target/test-classes/MyFirstJUnitJupiterTests.class

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        LabAssignment/Lab3_UnitTest/pom.xml
        LabAssignment/Lab3_UnitTest/src/
        LabAssignment/Lab3_UnitTest/target/
        LabDemo/Lab3_Unit/junit6-labdemo/target/maven-status/
        LabDemo/Lab3_Unit/junit6-labdemo/target/surefire-reports/

no changes added to commit (use "git add" and/or "git commit -a")

```

```

5
6 public class UnitTestCipher {
7
8     private final ShiftCipher cipher = new ShiftCipher();
9     //TCB1: message = "SOFTWARE", key = 3, expected result = "VRINZDUH"
10    @Test
11    void Test1() {
12        assertEquals("VRINZDUH", cipher.shift("SOFTWARE", 3));
13    }
14
15    @Test
16    void Test2() {
17        assertEquals("vriwzduh", cipher.shift("software", 3));
18    }
19
20    @Test
21    void Test3() {
22        assertEquals("TWFSLJ123", cipher.shift("ORANGE123", 5));
23    }
24
25    @Test
26    void Test4() {
27        assertEquals("Twfsjln", cipher.shift("Oranegi", 3));
28    }
29
30    @Test
31    void Test5() {
32        assertEquals("GfsfSF", cipher.shift("BAnaNA", 5));
33    }
34
35
36 }

```

```
tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git status
On branch 2026
Your branch is up to date with 'origin/2026'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   LabAssignment/Lab3_JUnitTest/src/test/java/sqa/lab/UnitTestCipher.java
        modified:   LabAssignment/Lab3_JUnitTest/target/test-classes/sqa/lab/UnitTestCipher.class

no changes added to commit (use "git add" and/or "git commit -a")

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git add .
warning: in the working copy of 'LabAssignment/Lab3_JUnitTest/src/test/java/sqa/lab/UnitTestCipher.java', LF
it touches it

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git commit -m "test case 1"
[2026 82c6207] test case 1
2 files changed, 17 insertions(+), 2 deletions(-)

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ git push
Enumerating objects: 29, done.
Counting objects: 100% (29/29), done.
Delta compression using up to 16 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (15/15), 1.40 KiB | 1.40 MiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
To https://github.com/tankeiei/673380301-0-SQA.git
   53ca8b8..82c6207 2026 -> 2026

tanku@Tanny-PC MINGW64 ~/Desktop/Lab3 QA/673380301-0-SQA (2026)
$ |
```